25901870 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Skill Development Components (SDC) - MG2SDCREM102 - MATHEMATICS FOR RENEWABLE ENERGY- 2

(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **15** questions

Each question carries **1** marks

1. Give another name of NOT GATE [U] / [CO1]
2. Draw the logic symbol of NOR GATE [U] / [CO1]
3. Which are the values taken by a boolean variable? [U] / [CO1]
4. Write the equation of the parabola with vertex (2,5) and open upwards [A] / [CO2]
5. What is the foci of the ellipse with centre (h,k) and major axis along x axis? [A] / [CO2]
6. What is the centre of the hyperbola $\frac{x^2}{16} - \frac{y^2}{25} = 1$? [A] / [CO2]
7. What is the distance between foci of an ellipse? [A] / [CO2]
8. Write the equation of the hyperbola with centre(-4,-8)and transverse axis is along x axis. [A] / [CO2]
9. What is the length of latus rectum of the hyperbola $\frac{x^2}{16} - \frac{y^2}{4} = 1$? [A] / [CO2]

10. In blending lpp the objective is often to : [C] / [CO3]

11. What is meant by maximization problem? [C] / [CO3]

12. A diet problem aims to meet nutritional requirement at : [C] / [CO3]

13. Define a cycle. [U] / [CO4]

14. Draw a 3- regular graph. [U] / [CO4]

15. Draw a graph and its sub graph. [U] / [CO4]

16. What is simple graph? [U] / [CO4]

17. Write the number of vertices of W_n ? [U] / [CO4]

[1x15 = 15]

Part B

Short Essay

Answer any **3** questions

Each question carries **5** marks

18. What values of boolean variables x and y satisfy xy= x+y [U] / [CO1]

19. Find the eccentricity of the ellipse with length of major axis 10 and focus (0,4) [A] / [CO2]

20. Maximise Z = – x + 2y [C] / [CO3]
    subject to the constraints:
    $x \geq 3$
    $x + y \geq 5$
    $x + 2y \geq 6$
    $y \geq 0$.

21. Draw a graph G and its complement [U] / [CO4]

22. Draw a complete graph K_5 and K_4 [U] / [CO4]

[5x3 = 15]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

23. Prepare a truth table for the boolean function F(x,y,z)=xy+xc [U] / [CO1]

24. Find the equation of the parabola with vertex (0,0) passing through (5,2) and symmetric to y axis [A] / [CO2]

25. A factory makes tennis rackets and cricket bats. A tennis racket takes 1.5 hours of machine time and 3 hours of craftman's time in its making while a cricket bat takes 3 hour of machine time and 1 hour of craftman's time. In a day, the factory has the availability of not more than 42 hours of machine time and 24 hours of craftsman's time .(i) What number of rackets and bats must be made if the factory is to work at full capacity? (ii) If the profit on a racket and on a bat is Rs 20 and Rs 10 respectively, find the maximum profit of the factory when it works at full capacity. [C] / [CO3]

26. Write a short note on different types of matrix representation. [U] / [CO4]

[10x2 = 20]